

Progress Report 2: Encoder Bandwidth, Not Foveation Specifically, Shapes Spatial Memory in PointNav Agents

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I. QUESTION AND HEADLINE FINDING

Question. A PointNav agent [1], [2] has two routes to spatial information its policy needs to act: recompute it each step from the current visual encoder output, or maintain it across time inside the recurrent (LSTM) memory. These routes compete for finite hidden-state capacity. *Does the structure of the visual input determine how this capacity gets allocated?*

Headline finding. Across five conditions that share architecture, sensor stack, training budget, and within-episode behaviour (96–99% success), the linear top-layer GPS R^2 falls *monotonically* with encoder spatial output dimension (Figure 1, solid line). The same hidden states recover GPS at $R^2 \in [0.46, 0.92]$ under a 2-layer MLP probe (Figure 1, dashed line) — so the contrast is about *linear* top-layer readability (the operation the DD-PPO action head performs), not information loss.

Theoretical claim. Bottleneck conditions force the LSTM to integrate the GPS sensor across time and store the result in a linearly-readable axis of $\mathbf{h}_2$ — there is no other route. A rich encoder offers a viable per-step visual-feature route; policy gradient reallocates capacity to it, and the integrated code is no longer the linearly-readable axis. The same hidden state encodes position in both regimes; only the format differs.

II. FIVE CONDITIONS ON ONE BANDWIDTH AXIS

All agents share a 3-layer LSTM (512-d) and the Wijmans [3] non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar); only the visual input changes. All four sighted conditions were retrained from scratch under unified hyperparameters (seed=0, num_envs=16, 250M frames), eliminating the hyperparameter inconsistency present in our *Report 1* runs. All achieve 96–99% success and SPL $\in [0.85, 0.89]$, so any cross-condition representational difference cannot be attributed to a skill gap.

Table I

FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT. BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\max} = 10$)
Uniform	4×4	256×256 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes ($\sim 30\text{k}$ – 120k LSTM steps) and run six analytical lenses on the top-layer $\mathbf{h}_2$: linear (Ridge $\alpha=10$) and MLP

(256-d, $L_2=10^{-4}$) probes for GPS; lag- k probe for $k \in \{0, 1, 2, 5, 10, 20, 50\}$; per-(unit, scene) Skaggs spatial information (rectified $\max(\mathbf{h}_2, 0)$); cross-training probes at five intermediate checkpoints; and cross-condition CKA, Procrustes shape distance, and memory transplant. All probes use 5-fold episode-level cross-validation. Single seed per condition follows the comparative-cognition modelling convention; rigour at the cross-condition level instead derives from convergence across the six lenses.

III. RESULTS: A BECAUSE-BUT-THEREFORE CHAIN

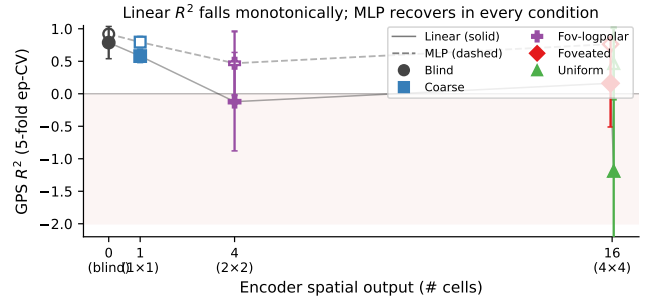


Figure 1. **The capacity-allocation principle, magnitude axis.** Top-layer GPS R^2 (5-fold ep-CV mean $\pm \sigma$) plotted against encoder spatial output. Solid line / filled markers: linear (Ridge) probe — monotonic anti-correlation with encoder bandwidth, dropping from +0.79 (blind) to -1.19 (uniform). Dashed line / open markers: 2-layer MLP probe on the *same* hidden states — recovery in every condition, gap widens with bandwidth. Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions a linear policy head cannot read. Blind row from preliminary 100-ep partial NPZ; full 500-ep collection in flight.

Because. Linear GPS R^2 falls monotonically with encoder spatial output (Figure 1, solid line). Bottleneck conditions retain the linear code (+0.79 blind, +0.58 coarse); rich-encoder conditions lose it (+0.16 foveated, -1.19 uniform). Foveated-logpolar with a 2×2 encoder sits at -0.12, which is the H1-mechanism prediction made *after* Report 1: a third rich-encoder condition with intermediate spatial output should land in the rich-encoder regime, and it does. Distance-to-goal — a same-dimensional control target — decodes at $R^2 \in [0.81, 0.96]$ in every condition, ruling out a probe-capacity-collapse reading.

But. A 2-layer MLP fit on the *same* hidden states recovers GPS in every condition (Figure 1, dashed line). The MLP-minus-linear gap widens monotonically with encoder bandwidth (+0.13 blind, +0.22 coarse, +0.59 fov-logpolar, +0.60 foveated, +1.65 uniform). Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions that the linear DD-PPO action head cannot read.

Therefore. The most parsimonious account is the capacity-allocation principle stated above: *integration route* (carry sensor across time) and *visual-feature route* (substitute encoder features for the integrated quantity) compete for finite recurrent capacity, and encoder bandwidth is the knob that determines which route wins. The substitution can be *policy-relevant* or *format-only*; Section IV disentangles these.

IV. CONVERGENT EVIDENCE

Three additional lenses corroborate the capacity-allocation account, each addressing an alternative explanation.

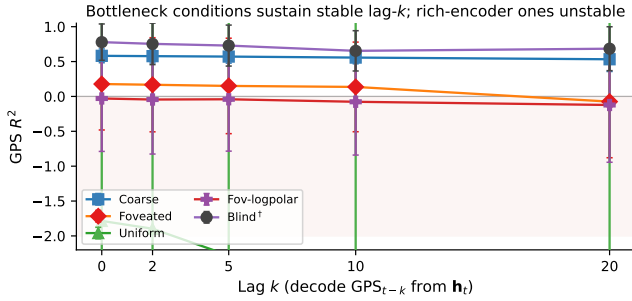


Figure 2. **Lag- k temporal stability separates integrated from reactive memory.** Decoding the past position GPS_{t-k} from the current $\mathbf{h}_t$. Bottleneck conditions (blind, coarse) sustain stable positive R^2 across the lag range; rich-encoder conditions (foveated, fov-logpolar, uniform) degrade rapidly. Bottleneck $\mathbf{h}_t$ is a smoothly-varying integrated code; rich-encoder $\mathbf{h}_t$ is reactive to current visual context. Blind† row from preliminary 100-ep partial NPZ.

Substitution dynamics across training. Re-probing intermediate checkpoints ($\approx 50/100/150/200/250$ M frames) shows that *all four sighted conditions start at high linear GPS R^2 at ~ 50 M frames* (coarse $+0.91$, foveated $+0.91$, uniform $+0.83$, foveated-logpolar $+0.92$); the linear top-layer code emerges early across the bandwidth spectrum. Bottleneck (coarse) preserves R^2 around $+0.5$ through training; rich-encoder conditions decay — uniform crosses zero by ~ 200 M and lands at -1.79 , foveated drifts to $+0.18$, foveated-logpolar to -0.03 . Decay rate tracks encoder informativeness about position — this is capacity allocation as a *training-time process*, not a property of the converged endpoint.

Lag- k temporal stability. Figure 2: bottleneck conditions sustain stable positive R^2 across $k \in \{0, 2, 5, 10, 20\}$; rich-encoder conditions are unstable. The *format* of the bottleneck-state code is a smoothly-varying integrated quantity; the rich-encoder state is reactive to the current visual scene and does not extrapolate temporally.

Place-cell signature is condition-flat. Per-(unit, scene) Skaggs spatial information is essentially condition-flat: 1.20 (coarse), 1.24 (foveated), 1.19 (uniform), 1.16 (foveated-logpolar); blind 1.35 (preliminary). $\Delta \approx 0.15$ bits, well within the within-condition spread. The bandwidth-allocation signal lives in the *linear-readout axis*, not in the per-unit place-cell metric. This is a defence against the obvious null reading (“maybe blind/coarse just have more place cells”): they do not.

V. WHAT CHANGED SINCE REPORT 1; STATUS; RISKS

Reframe. Report 1 reported foveated as having the longest temporal memory ($R^2=0.629$ at lag-5, the only condition still positive). This is *overturned*. The Report 1 ranking reflected partially-converged training plus heterogeneous hyperparameters across conditions; after unified retraining, foveated’s lag- k is *less* stable than coarse’s, not more (Figure 2). The new mechanism (encoder bandwidth) makes the same predictions for blind/coarse/uniform/foveated and additionally predicts that foveated-logpolar (a 2×2 rich-encoder control not in Report 1) lands in the rich-encoder regime — which it does (-0.12 , Figure 1).

Status. Phase A probe-collect: 4/5 complete; blind 500-ep run in flight on EPFL RCP (~ 6 h ETA, partial 100-ep NPZ already used in both figures above). Phase B (linear/MLP/lag- k /Skaggs/substitution/CKA-Procrustes-transplant) complete for the four sighted conditions; the blind row of each is queued to fire automatically once the 500-ep NPZ lands. Pre-DDL queue: re-run lenses with converged blind, apply values to the NeurIPS submission, sanity-rebuild the paper PDF.

Risks tracked. (a) The 50-ep blind preview gave catastrophically wide CV folds (linear -0.27 ± 1.41); at 100 eps the same probe stabilised at $+0.79 \pm 0.25$ (the value plotted). We expect 500 eps to tighten std to ~ 0.10 . (b) A separately-trained blind on a different seed/effective-batch gives a different mean ($+0.36$); we use our own retrained checkpoint and treat the alternative as one cross-setup robustness check.

Contributions. W. Xu post-retrain probe-collect, six-lens analysis pipeline, NeurIPS paper revision, this report. L. Bruneau foveation transform, training infrastructure, learned-gaze decoder, CKA/Procrustes pipelines. Z. Aoutir substitution-dynamics figure, place-cell visualisations. R. Abkari behavioural ablations (shortcut, GPS-zeroing, excursion-forgetting), MP3D evaluation.

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